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# Med-R1: Reinforcement Learning for Generalizable Medical Reasoning in Vision-Language Models

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# **Introduction:** Challenges of Current Medical VLMs

**Limitations of Supervised Fine-Tuning (SFT)**

- **Annotation Bottleneck**
  - Medical Chain-of-Thought (CoT) Data: High cost, expert-dependent
- **Shortcut Learning**
  - Models memorize superficial patterns instead of learning actual clinical logic
- **"Black-Box" Predictions**
  - SFT: Non-interpretable, low trust, non-compliant
- **Poor Generalization**
  - Performance gaps in diverse modalities & tasks

# **Introduction:** Med-R1: RL-Driven Reasoning

**The Proposed Framework: Reinforcement Learning (RL)**

- **Emergent Reasoning**
  - Autonomous reasoning via RL (Zero-human labels)
- **GRPO Algorithm**
  - Value-model-free training via GRPO (Group Relative Policy Optimization)
- **Key Benefits**
  - **Generalizability**
    - 8-modality robustness without retraining
  - **Trustworthiness**
    - Generates explicit, interpretable "Think" processes for every diagnosis
  - **Efficiency**
    - Enables a 2B model to outperform counterparts 36x its size

# **Method:** Mathematical Core: GRPO Objective

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q\sim P(Q),\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{odd}}(O|q)}$$
$$\frac{1}{G}\sum_{i=1}^{G}\left[min\left(\frac{\pi_{\theta_{new}}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, clip\left(\frac{\pi_{\theta_{new}}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right)\right.$$
$$\left. - \beta\mathbb{D}_{\text{KL}}\left(\pi_{\theta_{new}} \| \pi_{\theta_{ref}}\right)\right]$$

**GRPO Objective Function Key Mathematical Components**

- **Policy Ratio** $\frac{\pi_{\theta_{new}}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}$
  - Measures the change between the current policy and the old policy
- **Advantage Estimation** $A_i$
  - Normalized group rewards (vs. PPO's Value Model)

# **Method:** Mathematical Core: GRPO Objective

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$$\left. - \beta\mathbb{D}_{\text{KL}}\left(\pi_{\theta_{new}} \| \pi_{\theta_{ref}}\right)\right]$$

**GRPO Objective Function Key Mathematical Components**

- **Clipping Mechanism**
  - Restricts the policy update to a range $[1-\varepsilon, 1+\varepsilon]$ to ensure training stability
- **KL Divergence $\mathbb{D}_{KL}$**
  - Linguistic Capability Preservation

# **Method:** Reward Logic & Advantage Calculation

**GRPO: Group-Relative Advantage ($A_i$) Calculation**

- **Mechanism**
  - Generates $G$ independent responses per prompt to compute relative advantage
- **$A_i$ Calculation**
  - Normalizes reward $r_i$ using group mean and standard deviation:

$$A_i = \frac{r_i - \text{mean}(r_{1\ldots G})}{\text{std}(r_{1\ldots G})}$$

- **Efficiency**
  - Eliminates the Value Function $V(s)$, significantly reducing GPU memory overhead
- **Rule-Based Rewards ($r_i$)**
  - $R_{format}$: Binary for correct <think> and <answer> tag usage
  - $R_{accuracy}$: Binary for matching the ground truth MCQ option

# **Performance Upper Bound:** No-Think RL

| | No-Think RL fine-tuned Qwen2.5-VL-3B | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **CT** | 98.27 ± 0.45 | 74.77 ± 1.04 | 85.33 ± 1.70 | 37.61 ± 2.05 | 69.07 ± 2.53 | 78.87 ± 2.46 | 87.26 ± 2.30 | 67.75 ± 2.75 | 74.87 ± 0.64 |
| **MRI** | 69.70 ± 1.56 | 99.61 ± 0.15 | 81.73 ± 1.89 | 34.91 ± 2.03 | 75.19 ± 2.34 | 80.51 ± 2.32 | 89.98 ± 2.00 | 67.48 ± 2.75 | 74.89 ± 0.65 |
| **X-Ray** | 80.10 ± 1.36 | 69.78 ± 1.14 | 95.79 ± 0.99 | 34.33 ± 2.05 | 71.29 ± 2.41 | 81.79 ± 2.28 | 83.14 ± 2.54 | 68.65 ± 2.79 | 73.11 ± 0.63 |
| **Ultrasound** | 56.40 ± 1.70 | 64.93 ± 1.18 | 80.25 ± 1.92 | 100.0 ± 0.00 | 68.38 ± 2.53 | 75.32 ± 2.60 | 76.30 ± 2.89 | 71.71 ± 2.66 | 74.16 ± 0.66 |
| **Dermoscopy** | 55.45 ± 1.70 | 66.48 ± 1.15 | 80.37 ± 1.95 | 37.42 ± 2.12 | 92.88 ± 1.38 | 71.86 ± 2.69 | 76.53 ± 2.83 | 67.12 ± 2.70 | 68.51 ± 0.69 |
| **Fundus** | 58.65 ± 1.68 | 66.08 ± 1.16 | 82.79 ± 1.83 | 34.33 ± 2.10 | 68.68 ± 2.49 | 93.81 ± 1.41 | 79.72 ± 2.71 | 65.86 ± 2.79 | 68.74 ± 0.69 |
| **OCT** | 80.38 ± 1.36 | 75.02 ± 1.05 | 86.81 ± 1.67 | 36.11 ± 2.05 | 72.21 ± 2.41 | 81.60 ± 2.28 | 99.88 ± 0.18 | 65.32 ± 2.79 | 74.67 ± 0.64 |
| **Microscopy** | 60.17 ± 1.73 | 66.59 ± 1.16 | 83.10 ± 1.86 | 36.60 ± 2.10 | 71.29 ± 2.45 | 77.05 ± 2.46 | 80.66 ± 2.71 | 97.66 ± 0.86 | 71.64 ± 0.67 |
| **Overall** | 69.89 ± 1.59 | 72.91 ± 1.08 | 84.52 ± 1.73 | 43.91 ± 2.14 | 73.62 ± 2.37 | 80.10 ± 2.41 | 84.18 ± 2.47 | 71.44 ± 2.65 | 72.57 ± 0.66 |

1

2

1 **In-Domain Mastery (Diagonal)**

- Achieving near-perfect scores
- Maximum capacity: Raw performance without reasoning overhead

2 **System-Wide Benchmark**

- Theoretical Max: Unconstrained medical feature extraction

# **Result:** Cross-Strategy Performance Comparison

**Model Architecture: Qwen2.5-VL-3B**

- **No-Think RL (72.57%)**
  - Performance Ceiling
  - maximum feature-to-answer mapping
- **Think RL (71.14%)**
  - Hybrid Baseline
  - Rationalization occurs after the decision
- **Think-after RL (71.62%)**
  - Clinical Standard
  - Prioritizes "Thinking" before "Answering"
- **Technical Trade-off**
  - Transparency Cost: Negligible (~ 1.4%) accuracy penalty

# **Result:** Why "Think-After" Wins

- **Efficient Convergence:** Think-After > Think in speed and reward
- **Optimization:** Decouples decision-making from rationalization
- **Less Interference:** Pre-answer reasoning can disrupt autoregressive generation
- **Medical Reliability:** Minimizes hallucination while preserving interpretability

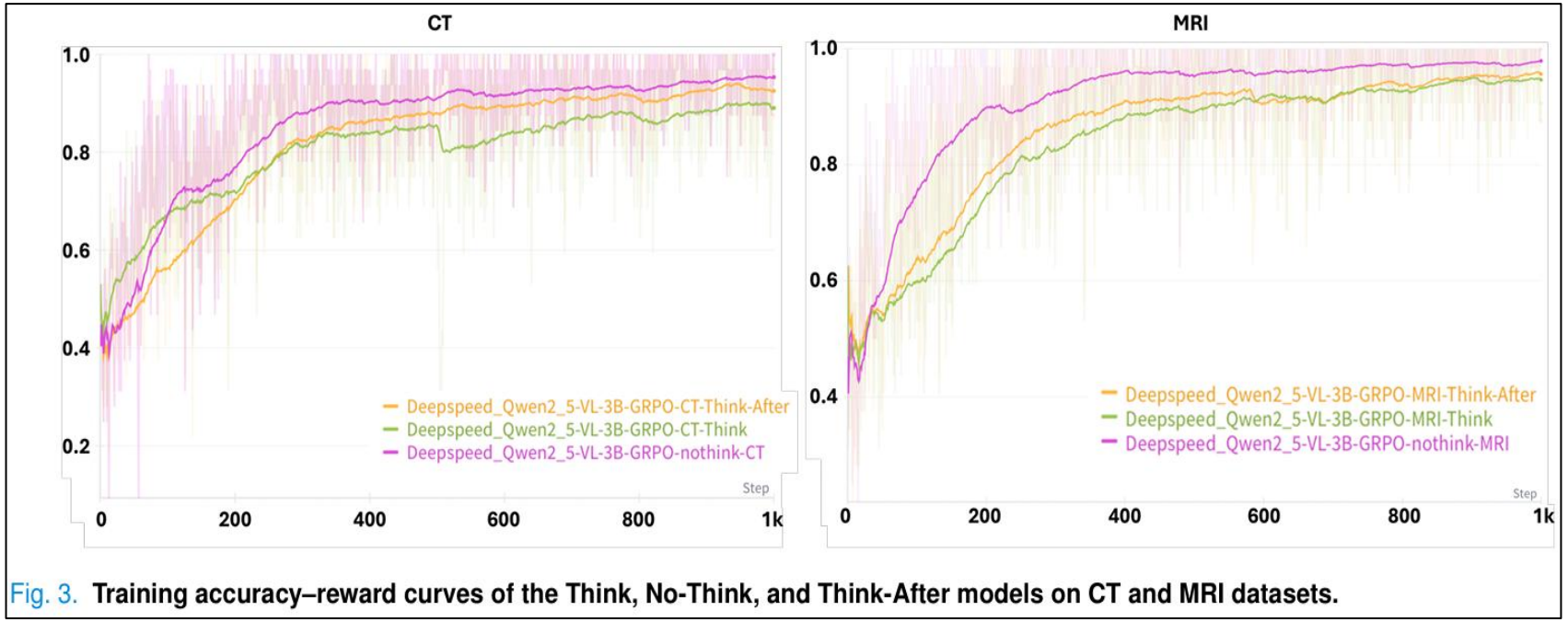



Fig. 3. **Training accuracy–reward curves of the Think, No-Think, and Think-After models on CT and MRI datasets.**

# **Consistency Analysis:** Anchor Modalities

**High-Impact Data Sources**

- **Training Leaders (Row Overall)**
  - **MRI (> 74%) & CT (> 72%)**
    - Most robust feature sources for any RL strategy
  - Anatomical density drives cross-modality generalization
- **Testing Targets (Column Overall)**
  - **OCT (> 82%) & X-Ray (> 83%)**
    - Most "receivable" modalities for zero-shot models
  - Cross-Source Feature Consistency

# **The Bottlenecks:** Generalization Gaps

**Universal Challenges for VLM Reasoning**

- **The Ultrasound Paradox**
  - 100% In-Domain (No-Think) vs. ~45% Average Transfer
  - Persistent noise overfitting despite Reasoning
- **The "Micro-to-Macro" Barrier**
  - Domain Gap: Microscopy features fail to generalize to CT and MRI
  - Domain mismatch: Micro-cellular textures vs. macro-structural anatomy

# **Qualitative Analysis:** Case Study

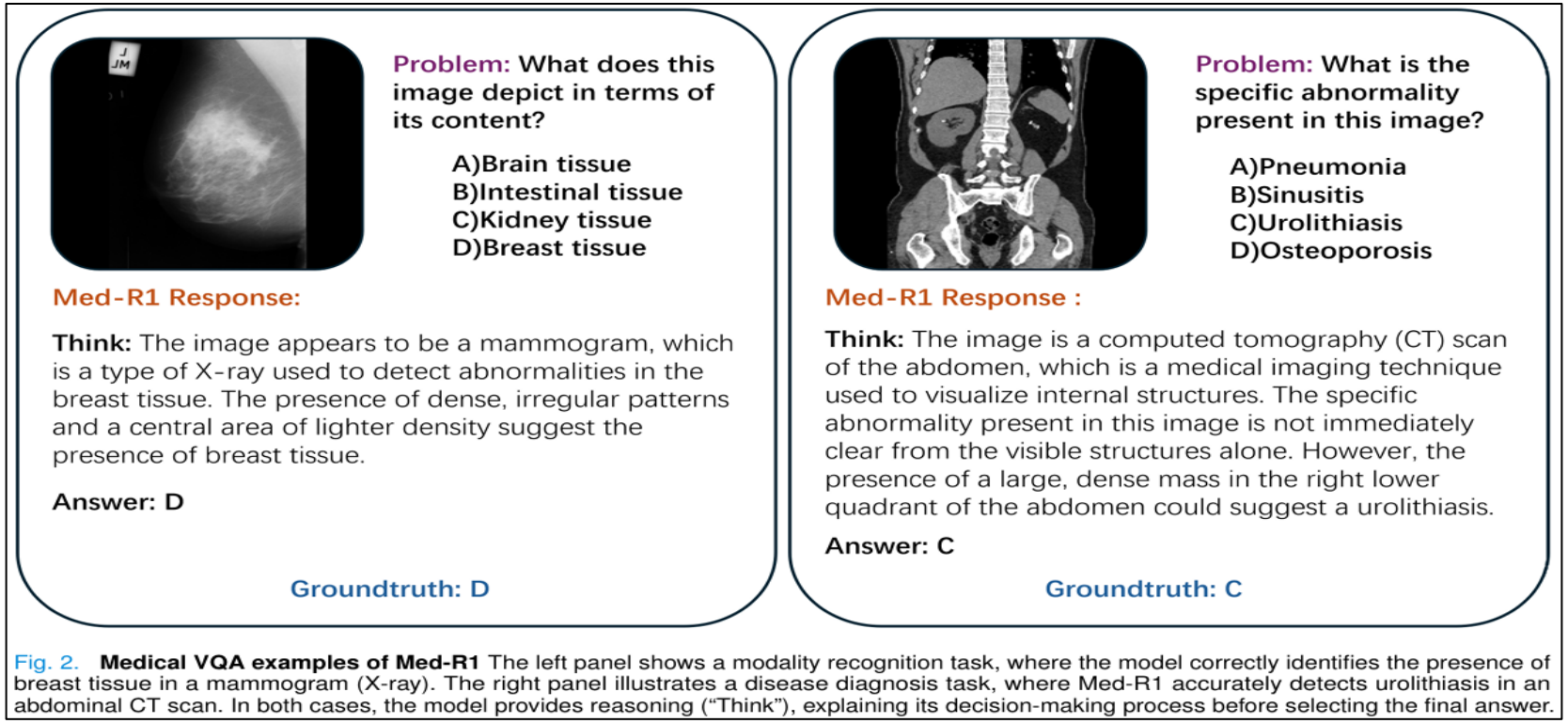



Fig. 2. **Medical VQA examples of Med-R1** The left panel shows a modality recognition task, where the model correctly identifies the presence of breast tissue in a mammogram (X-ray). The right panel illustrates a disease diagnosis task, where Med-R1 accurately detects urolithiasis in an abdominal CT scan. In both cases, the model provides reasoning ("Think"), explaining its decision-making process before selecting the final answer.

- **Traceable Logic:** Shows `<think>` process from visual features to diagnosis
- **Expert Alignment:** Reasoning paths follow clinical standards
- **Error Correction:** RL helps the model refine its path through trial and error

# **Clinical Validation:** Reader Study

TABLE V

**READER STUDY RESULTS.** THREE READERS INDEPENDENTLY EVALUATED 100 CORRECTLY ANSWERED VQA SAMPLES FOR FACTUAL CORRECTNESS (QUALITY, 1–5) AND REASONING–ANSWER CONSISTENCY (CONSISTENCY, %).

| Reader | Quality | Consistency (%) |
|---|---|---|
| A | 4.23 | 93.0 |
| B | 4.19 | 92.0 |
| C | 4.34 | 95.0 |
| **Mean** | **4.25** 1 | **93.3** 2 |

Quality reflects factual and clinical correctness of model answers; Consistency measures logical alignment between reasoning and answers.

1 **Factual Quality:** 4.25 / 5.0 (Rated by medical experts)
2 **Logic Consistency:** 93.3% alignment between reasoning and answers
3 **Verdict:** Proves Med-R1 provides trustworthy, human-readable rationales

# Conclusion

## Core Contributions

- **Scalable RL Framework**
  - Proves GRPO enables 3B models to surpass 72B
- **Paradigm Shift**
  - Think-After decouples decision from reasoning
- **Generalization Leader**
  - New benchmark for cross-modality reasoning

## Future Directions

- **From 2D to 3D or Video**
  - Supporting Volumetric CT and MRI and Ultrasound
- **Clinical Context**
  - Incorporating patient history and EHR data

# Discussion & Limitations

- **Reasoning Quality**
  - "More thinking" ≠ "Better performance" in medical VQA
  - Unsupervised rationales can introduce noise under domain shift
- **Dimension Constraint**
  - Currently frame-level (2D) only
  - Needs expansion to 3D Volumetric (CT, MRI) and Video
- **Data Bottleneck**
  - Scarcity of high-quality, expert-annotated CoT data

# My Insights & Critique

- **Accuracy vs. Reasoning Gap**
  - Accuracy vs. Reasoning Gap
    - Reasoning does not always improve prediction
- **Post-hoc Explanation Risk**
  - Think-After: Prediction → Reasoning
    - Explanation may not reflect true reasoning
- **Dataset Limitation: OmniMedVQA**
  - Mostly MCQ-style VQA tasks focus on recognition rather than diagnosis
- **Future Direction**
  - More complex clinical tasks
  - Integration with patient context / EHR

# The OmniMedVQA Dataset Sample

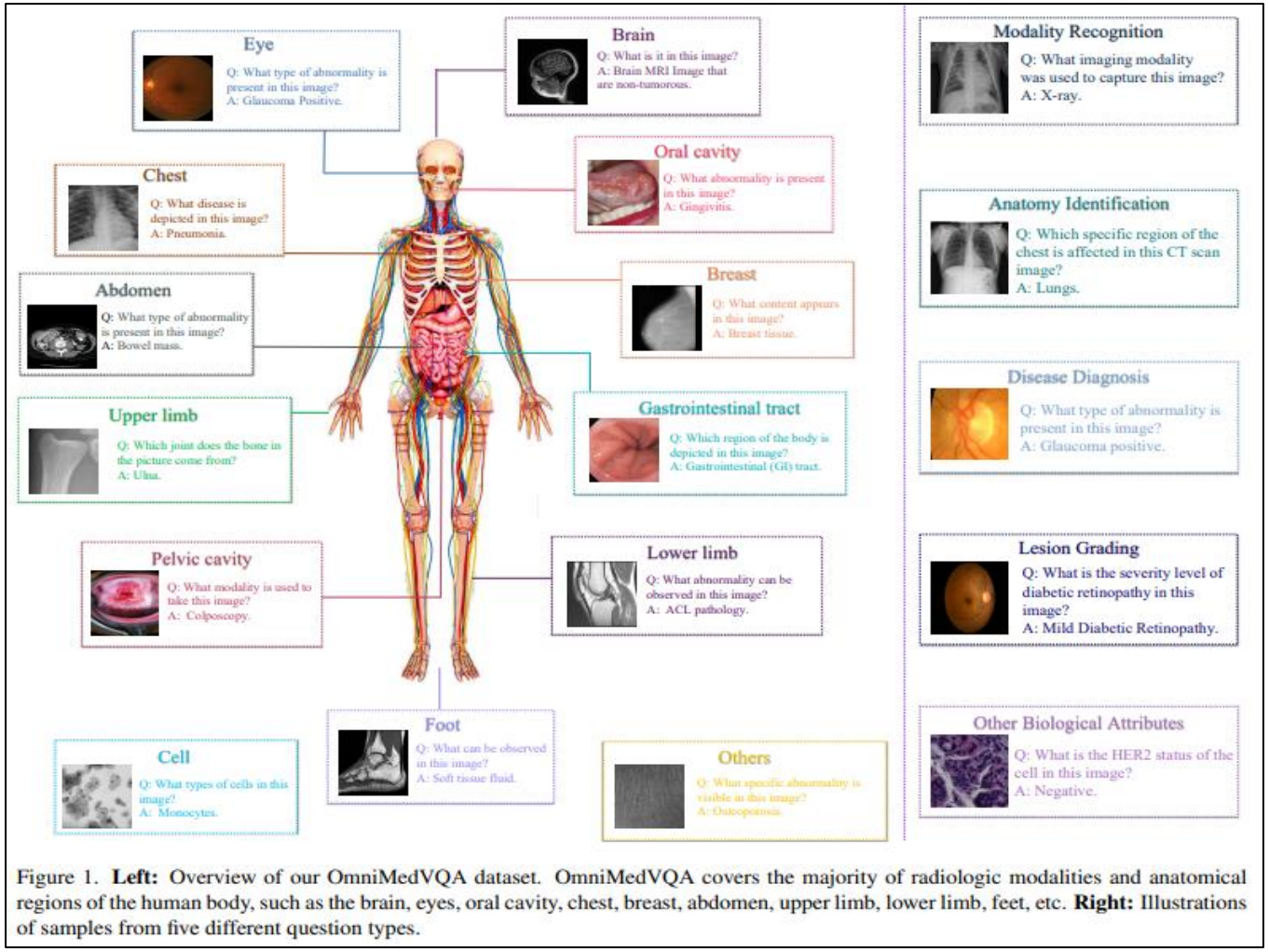



Figure 1. **Left:** Overview of our OmniMedVQA dataset. OmniMedVQA covers the majority of radiologic modalities and anatomical regions of the human body, such as the brain, eyes, oral cavity, chest, breast, abdomen, upper limb, lower limb, feet, etc. **Right:** Illustrations of samples from five different question types.

Y. Hu, T. Li, Q. Lu, W. Shao, J. He, Y. Qiao, and P. Luo, "OmniMedVQA: A New Large-Scale Comprehensive Evaluation Benchmark for Medical LVLM," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 22170–22183